```
!pip install streamlit
import streamlit as st
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import statsmodels.api as sm
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
```

```
Collecting streamlit
  Downloading streamlit-1.54.0-py3-none-any.whl.metadata (9.8 kB)
Requirement already satisfied: altair!=5.4.0,!=5.4.1,<7,>=4.0 in /usr/local/lib/
Requirement already satisfied: blinker<2,>=1.5.0 in /usr/local/lib/python3.12/di
Requirement already satisfied: cachetools<7,>=5.5 in /usr/local/lib/python3.12/d
Requirement already satisfied: click<9,>=7.0 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: gitpython!=3.1.19,<4,>=3.0.7 in /usr/local/lib/py
Requirement already satisfied: numpy<3,>=1.23 in /usr/local/lib/python3.12/dist-
Requirement already satisfied: packaging>=20 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: pandas<3,>=1.4.0 in /usr/local/lib/python3.12/dis
Requirement already satisfied: pillow<13,>=7.1.0 in /usr/local/lib/python3.12/di
Collecting pydeck<1,>=0.8.0b4 (from streamlit)
  Downloading pydeck-0.9.1-py2.py3-none-any.whl.metadata (4.1 kB)
Requirement already satisfied: protobuf<7,>=3.20 in /usr/local/lib/python3.12/di
Requirement already satisfied: pyarrow>=7.0 in /usr/local/lib/python3.12/dist-pa
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Requirement already satisfied: tenacity<10,>=8.1.0 in /usr/local/lib/python3.12/
Requirement already satisfied: toml<2,>=0.10.1 in /usr/local/lib/python3.12/dist
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Requirement already satisfied: typing-extensions<5,>=4.10.0 in /usr/local/lib/py
Requirement already satisfied: watchdog<7,>=2.1.5 in /usr/local/lib/python3.12/d
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages
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Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-pa
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Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/d
Requirement already satisfied: smmap<6,>=3.0.1 in /usr/local/lib/python3.12/dist
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist
Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/loca
Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/python3.12/
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Installing collected packages: pydeck, streamlit
```

```
Successfully installed pydeck-0.9.1 streamlit-1.54.0
```

```
# ============================================================
# KONFIGURASI HALAMAN
# ============================================================
st.set_page_config(page_title="Dashboard Kepuasan Pegawai", layout="wide")
st.title("📊 Dashboard Mini Kepuasan Layanan Kepegawaian")
st.markdown("Analisis berbasis data untuk mendukung rekomendasi kebijakan")
```

```
2026-03-02 10:17:06.503 WARNING streamlit.runtime.scriptrunner_utils.script_run_
2026-03-02 10:17:06.505 WARNING streamlit.runtime.scriptrunner_utils.script_run_
2026-03-02 10:17:06.670
  Warning: to view this Streamlit app on a browser, run it with the following
  command:

    streamlit run /usr/local/lib/python3.12/dist-packages/colab_kernel_launcher.
2026-03-02 10:17:06.671 Thread 'MainThread': missing ScriptRunContext! This warn
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2026-03-02 10:17:06.673 Thread 'MainThread': missing ScriptRunContext! This warn
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DeltaGenerator()
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```
# ============================================================
# LOAD DATA
# ============================================================
df = pd.read_excel("/content/data_simulasi_50_siswa_20_soal (2).xlsx")
# Ambil indikator V1-V5 dan pastikan numerik
indikator = df.iloc[:, 1:6].apply(pd.to_numeric, errors="coerce")
```

```
# ============================================================
# KPI KEPUASAN (IKM)
# ============================================================
mean_scores = indikator.mean()
ikm = (mean_scores.mean() / 5) * 100

def kategori_ikm(x):
    if x >= 81: return "Sangat Baik"
    elif x >= 66: return "Baik"
    elif x >= 51: return "Cukup"
    else: return "Kurang"

col1, col2, col3 = st.columns(3)
col1.metric("📈 Indeks Kepuasan (IKM)", f"{ikm:.2f}%")
col2.metric("🏷️ Kategori", kategori_ikm(ikm))
col3.metric("👥 Responden", len(df))

st.divider()
```

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DeltaGenerator()
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```
# ============================================================
# ANALISIS GAP
# ============================================================
st.header("3️⃣ Analisis GAP (Expectation vs Performance)")

gap_scores = 5 - mean_scores
prioritas_gap = gap_scores.idxmax()

fig_gap, ax_gap = plt.subplots(figsize=(6,4))
ax_gap.bar(gap_scores.index, gap_scores.values)
ax_gap.set_ylabel("Nilai GAP")
ax_gap.set_title("GAP Kepuasan per Indikator")
ax_gap.grid(axis="y", linestyle="--", alpha=0.6)

for i, v in enumerate(gap_scores.values):
    ax_gap.text(i, v + 0.03, f"{v:.2f}", ha="center", fontweight="bold")

st.pyplot(fig_gap)
st.info(f"📌 Prioritas perbaikan tercepat: **{prioritas_gap}**")

st.divider()
```

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DeltaGenerator()
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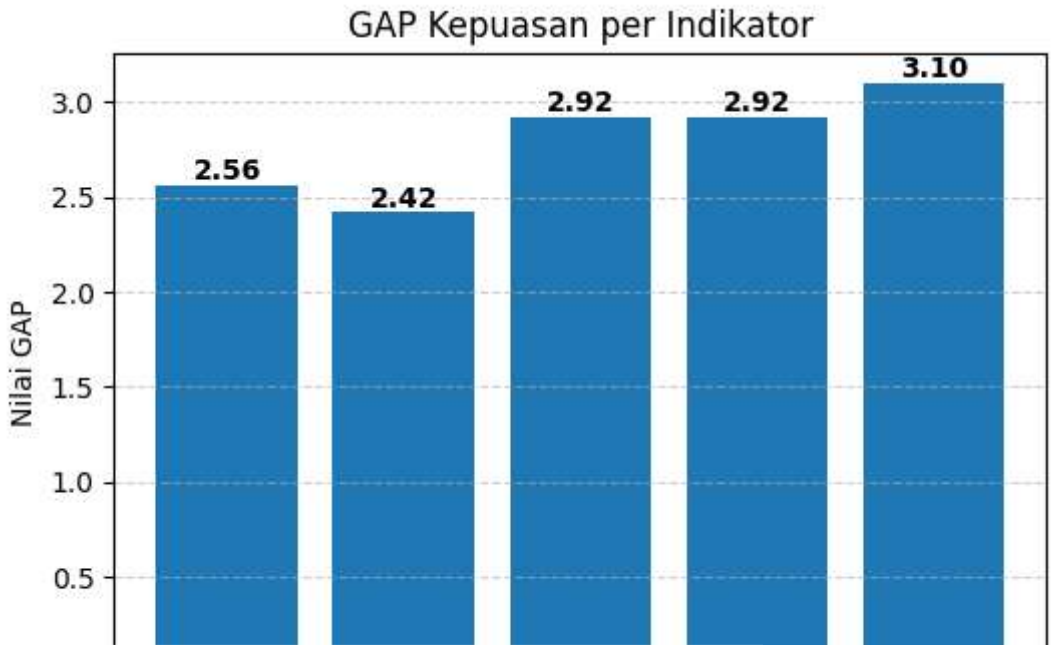


```
# ============================================================
# ANALISIS KORELASI
# ============================================================
st.header("4️⃣ Korelasi Antar Indikator")

corr = indikator.corr()

fig_corr, ax_corr = plt.subplots(figsize=(6,5))
im = ax_corr.imshow(corr, vmin=-1, vmax=1)
plt.colorbar(im, ax=ax_corr)

ax_corr.set_xticks(range(len(corr.columns)))
ax_corr.set_yticks(range(len(corr.columns)))
ax_corr.set_xticklabels(corr.columns, rotation=45, ha="right")
ax_corr.set_yticklabels(corr.columns)

for i in range(len(corr)):
    for j in range(len(corr)):
        ax_corr.text(j, i, f"{corr.iloc[i, j]:.2f}", ha="center", va="center")
```

```
ax_corr.set_title("Heatmap Korelasi Pearson")
st.pyplot(fig_corr)

corr_v5 = corr.iloc[:-1, -1].sort_values(ascending=False)
st.subheader("📊 Ranking Faktor Berpengaruh")
st.dataframe(corr_v5.to_frame("Koefisien Korelasi"))

st.divider()
```

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DeltaGenerator()
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```
# ============================================================
# ANALISIS REGRESI
# ============================================================
st.header("5️⃣ Analisis Regresi Linear Berganda")

X = sm.add_constant(indikator.iloc[:, 0:4])
y = indikator.iloc[:, 4]

model = sm.OLS(y, X, missing="drop").fit()

coef = model.params[1:]
r2 = model.rsquared

fig_reg, ax_reg = plt.subplots(figsize=(6,4))
ax_reg.bar(coef.index, coef.values)
ax_reg.axhline(0, linestyle="--")
ax_reg.set_title("Koefisien Regresi")

st.pyplot(fig_reg)
st.info(f"📈 Nilai R²: **{r2:.2f}**")
st.success(f"🔑 Faktor dominan: **{coef.abs().idxmax()}**")
```

```
st.divider()
```

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DeltaGenerator()
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```
# ========================================================
# SEGMENTASI KEPUASAN
# ========================================================
st.header("6️⃣ Segmentasi Kepuasan Pegawai")

scaler = StandardScaler()
X_scaled = scaler.fit_transform(indikator.fillna(indikat

kmeans = KMeans(n_clusters=3, random_state=42, n_init=10
cluster_label = kmeans.fit_predict(X_scaled)

indikator_cluster = indikator.copy()
indikator_cluster["Cluster"] = cluster_label

cluster_mean = indikator_cluster.groupby("Cluster").mean
cluster_mean = cluster_mean.sort_values(by=indikator.colu

cluster_mean["Segment"] = ["Sangat Puas", "Cukup Puas", '

labels = indikator.columns.tolist()
angles = np.linspace(0, 2*np.pi, len(labels), endpoint=Fa
angles += angles[:1]

fig_rad = plt.figure(figsize=(6,6))
ax_rad = plt.subplot(polar=True)

for i, row in cluster_mean.iterrows():
    values = row[labels].tolist() + [row[labels].tolist(
    ax_rad.plot(angles, values, label=row["Segment"])
    ax_rad.fill(angles, values, alpha=0.25)
```

```
ax_rad.set_thetagrids(np.degrees(angles[:-1]), labels)
ax_rad.set_ylim(0, 5)
ax_rad.set_title("Radar Segmentasi Kepuasan")
ax_rad.legend(loc="upper right")

st.pyplot(fig_rad)

st.success("📌 Segmentasi berhasil – siap untuk rekomend
```

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DeltaGenerator()
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